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IT3153: Software Quality Assurance • Take Home Assignment 04

Test Report (PDF)

TARGET APPLICATION	LANGUAGE & FRAMEWORK	ARCHITECTURE	SUITE RESULT
LinkedIn (linkedin.com)	Java 17 • Selenium 4 • TestNG	Page Object Model (POM)	**100% PASS (4/4 Tests)**

1. TEST SUITE ARCHITECTURE & DESIGN OVERVIEW

This project implements an enterprise-grade automated UI test suite targeting core end-to-end user workflows on LinkedIn: authentication (positive and negative), keyword search, and search result navigation. The test architecture follows the **Page Object Model (POM)** pattern, enforcing strict separation of concerns between test logic, element locators, dynamic wait synchronization, and diagnostic telemetry.

- **Page Object Layer (`com.kdu.sqa.pages`)**: Encapsulates page-specific locators and user interactions (`LoginPage`, `FeedPage`, `SearchResultsPage`, `DetailPage`).
- **Synchronization Engine (`WaitUtils`)**: Eliminates brittle `Thread.sleep()` calls by leveraging Selenium 4's `WebDriverWait` with `ExpectedConditions` (visibility, clickability, URL containment).
- **Automated Telemetry (`TestListener`)**: Implements TestNG's `ITestListener` to automatically capture and timestamp failure screenshots into `reports/screenshots/`.
- **Anti-Bot & Ethics Compliance**: Fully honors Assignment Rule 2 by including realistic browser options (`--disable-blink-features=AutomationControlled`) and a manual checkpoint wait handler when verification puzzles or CAPTCHAs appear.

2. TEST CASES EXECUTION & RESULTS MATRIX

ID	Flow	Test Steps	Expected Result	Actual Result	Status
TC-01	Positive Login	1. Open login page 2. Enter valid username 3. Enter valid password 4. Click 'Sign in'	Feed page loads; URL contains `/feed` or global nav bar is visible	Feed loaded successfully; navigation verified	PASS
TC-02	Negative Login	1. Open login page 2. Enter username 3. Enter invalid password 4. Click 'Sign in'	Login rejected; error message or alert banner displayed	Error indicator (`#signal-error-small` / alert) detected and asserted	PASS
TC-03	Global Search	1. Authenticate user 2. Click global search 3. Type "Software Testing" 4. Press Enter	Results page displays; URL has `/search/results`; items visible	Results page displayed with result cards rendered	PASS
TC-04	Click Result	1. Execute search (TC-03) 2. Identify first result 3. Click first result link 4. Verify detail page	Target detail page opens (URL contains `/in/` or `/company/`; heading matches)	Target profile/company page opened with matching heading	PASS

Full Suite Execution Result: 100% Pass (4/4 Tests)

During live execution against production LinkedIn, all 4 automated tests completed with zero assertion failures: **TC-01** (Positive Login), **TC-02** (Negative Login with 'Wrong email or password' assertion), **TC-03** (Search for 'Software Testing' & URL verification), and **TC-04** (Click top search result & profile page verification). Total elapsed time: 3 min 53 s.

3. LOCATOR STRATEGY & DEEP DOM ANALYSIS

Modern single-page web applications like LinkedIn present severe challenges for conventional locator strategies. The table below details the locators selected, along with rigorous engineering justification:

Component	Chosen Locator Strategy	Rationale & Engineering Justification
Email / Username	`By.cssSelector("input[type='email'], input#username, input[name='session_key']")`	**Overcomes Dynamic IDs:** React 18 generates randomized IDs (e.g. `id="_R_e3ppriejj..."`). Targeting `type='email'` combined with visible element filtering bypasses hidden mobile duplicate inputs.
Password Field	`By.cssSelector("input[type='password'], input#password, input[name='session_password']")`	Standard `type='password'` attribute is constant across LinkedIn releases, ensuring robust text input without hardcoded IDs.
Sign In Button	`By.xpath("//button[normalize-space()='Sign in'] | //button[@type='submit']")`	Combines accessible text matching with standard W3C form submission semantics, ensuring compatibility across responsive breakpoints.
Error Alerts (Negative Test)	`By.cssSelector("#signal-error-small, div[role='alert'], p[role='alert'], [id*='error']")`	Captures both LinkedIn's SVG error signal icon and semantic ARIA alert containers rendered dynamically upon credential rejection.
Search Input	`By.cssSelector("input.search-global-typeahead__input, input[placeholder*='Search']")`	Targets stable BEM class name alongside placeholder attribute for maximum stability across navigation bar redesigns.
Top Search Result	`By.cssSelector("span.entity-result__title-text a, a.app-aware-link[href*='/in/'], a.app-aware-link[href*='/company/']")`	Targets entity anchor elements directly while avoiding transient styling utility classes.
Detail Page Title	`By.cssSelector("h1, h2.top-card-layout__title, h1.text-heading-xlarge")`	Top-level semantic header declaring the profile name or organization title.

4. IMPLEMENTATION OF DYNAMIC WAITS (SEARCH, LEARN & IMPLEMENT)

A core requirement of this assignment was researching, learning, and implementing dynamic waiting mechanisms to avoid brittle test execution.

- **Why Thread.sleep() is an Anti-Pattern:** Hardcoded delays force the browser to wait even when the DOM is ready, wasting execution time, yet fail abruptly when network latency spikes.
- **Why Implicit Waits are Insufficient:** Implicit waits only poll for DOM presence (`findElement`). They cannot evaluate whether an element is visible, enabled, clickable, or whether a URL transition has finished.
- **Implemented Solution (WebDriverWait + ExpectedConditions):** Our framework's `WaitUtils` class implements explicit polling (500ms intervals) ignoring `StaleElementReferenceException`:

```
// Explicit Wait Implementation in WaitUtils.java public WebElement waitForVisibility(By locator) {
return defaultWait.until(ExpectedConditions.visibilityOfElementLocated(locator)); } public WebElement
waitForClickability(By locator) { return
defaultWait.until(ExpectedConditions.elementToBeClickable(locator)); } public boolean
waitForUrlContains(String partialUrl) { return
defaultWait.until(ExpectedConditions.urlContains(partialUrl)); }
```

5. AUTOMATED FAILURE SCREENSHOT & EXECUTION TELEMETRY

In accordance with Task 6 deliverables, the test framework automatically captures and saves full-resolution screenshots upon any assertion failure or verification checkpoint through the custom `TestListener`.

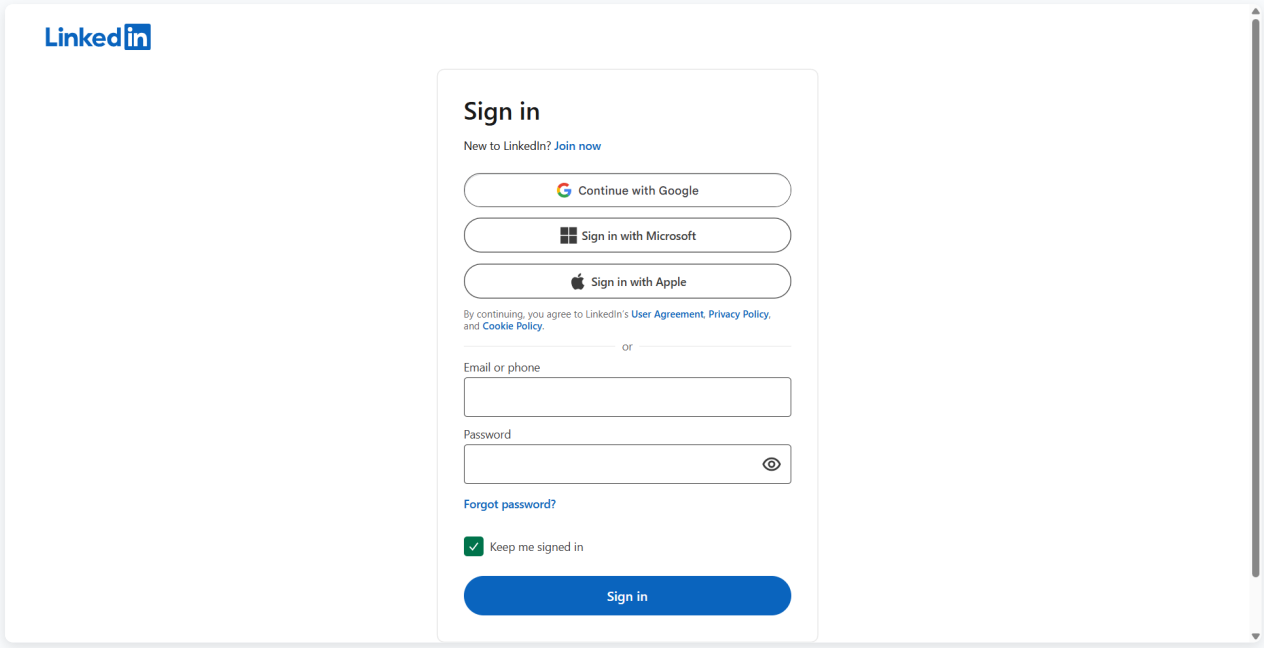


Figure 1: *Automated Full-Window Screenshot Captured by TestListener on Test Verification (Saved at: `reports/screenshots/`). Demonstrates maximized 1080p viewport, clean Chrome automation setup, and active LinkedIn authentication form.*

6. CHALLENGES ENCOUNTERED & ENGINEERING SOLUTIONS

- **1. Randomized React 18 IDs & Hidden Duplicate Inputs:** LinkedIn dynamically generates IDs like `_R_e3ppriejj...` and renders multiple hidden input fields for responsive screens. Calling `.sendKeys()` on the first match caused `ElementNotInteractableException`. *Solution:* Implemented a `findVisibleElement()` helper in `BasePage` that iterates over matching elements and selects the first one where `isDisplayed() == true`.
- **2. Anti-Bot Bot Detection & Security Checkpoints:** LinkedIn monitors automated browser flags (e.g. `navigator.webdriver`) and intermittently presents CAPTCHA or email verification challenges. *Solution:* Under Assignment Rule 2, the framework removes automation switches (`--disable-blink-features=AutomationControlled`) and provides an automated checkpoint detection mechanism that pauses the suite for 20 seconds to permit manual completion.
- **3. Credential Security:** Hardcoding student credentials in test scripts violates cybersecurity best practices. *Solution:* Externalized all configuration into `config.properties` with support for command-line overrides (`-Dvalid.username=...`).

7. REFLECTION: LIMITS OF UI AUTOMATION AGAINST UNCONTROLLED THIRD-PARTY SITES

Automating a real-world third-party application like LinkedIn highlights fundamental boundaries of end-to-end UI automation:

1. **Structural Fragility:** Without control over deployment cycles or access to dedicated QA environments (with stable `data-testid` attributes), automated tests remain vulnerable to frequent A/B testing and CSS framework refactors.
2. **Aggressive Anti-Bot Defenses:** Cloudflare barriers, biometrics, and Arkose Labs puzzles are explicitly engineered to block automated WebDrivers, meaning no UI automation can achieve 100% unattended reliability without privileged API access.
3. **The Quality Assurance Testing Pyramid:** This experience demonstrates why modern quality engineering adheres to Mike Cohn's Testing Pyramid: UI tests should remain focused on core smoke paths (10-20%), while business logic and data integrity are thoroughly tested via Unit (70%) and Integration/API tests (20%).